

A.19 Triple-Difference Including the Largest Firms

As described in Section 4.1, our main sample excludes the approximately 800 largest firms in the Revelio data (ranked by post-2021 new hires) to prevent the estimates from being driven by a handful of extreme outliers. In this appendix, we re-estimate the triple-difference specification of Equation 11 on a sample that adds these firms back. Acquiring and processing the full raw job-postings corpus for these firms—needed to classify their postings as GenAI integrator vacancies—is computationally prohibitive given the scale of their posting volume. However, because these firms are very likely to qualify as adopters under our definition given their size and posting activity, we include them in this robustness check as adopters.

Figure A.23 reports the resulting coefficients. The triple-difference profile is essentially unchanged from the baseline in Figure 5: the coefficients are flat through 2022Q4—consistent with parallel pre-trends—and then decline sharply beginning in 2023Q1, reaching roughly 9 percent by 2025Q1. Including the largest firms therefore does not alter the seniority-biased contraction in junior employment relative to senior employment documented in the main analysis.

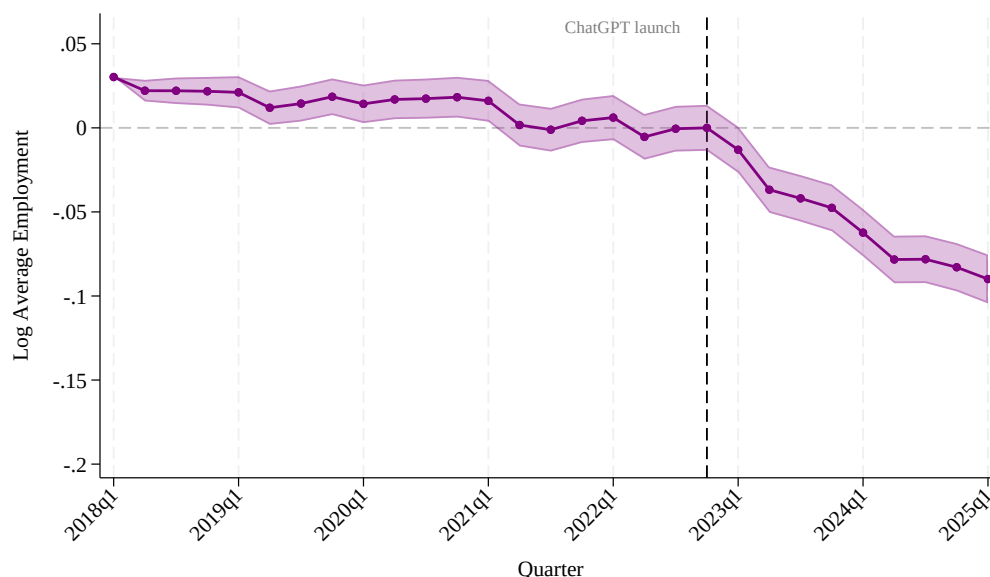


Figure A.23: Triple-Difference Including the Largest Firms

Notes: This figure reproduces the triple-difference of Equation 11 (Figure 5) on the main analysis sample augmented with the approximately 800 largest firms (ranked by post-2021 new hires) that are excluded from the main analysis. Because their firm-level adoption date is poorly identified, these firms are treated as GenAI adopters. The coefficients capture how junior employment evolves relative to senior employment within the same firm and period, comparing adopters to non-adopters, and are normalized to zero in 2022Q4. Standard errors are clustered at the firm level.

A.20 Mentions of “AI” on Earnings Calls

Figure A.24 shows that mentions of “AI” on U.S. firms’ earnings calls began rising steeply as early as 2022Q4, tripling by 2023Q2. This pattern is referenced in Section 5.2 as evidence that firms were aware of and responding to the diffusion of GenAI almost immediately after the release of ChatGPT, consistent with the forward-looking interpretation of the junior employment decline.

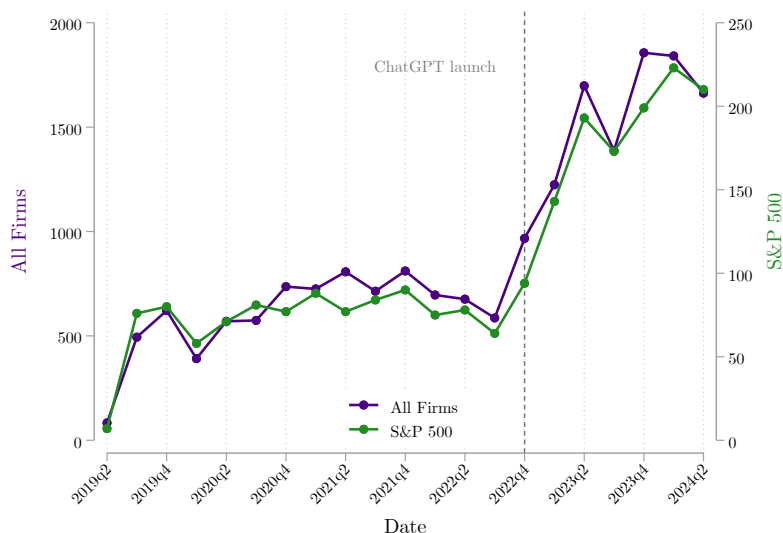


Figure A.24: Mentions of “AI” on Earnings Calls

Notes: Reproduced from Sherwood News (2024), using data from FactSet CallStreet earnings transcripts (FactSet Research Systems Inc., 2025).

A.21 Early 2023 Labor-Market Adjustments to GenAI—Anecdotal Evidence

In this appendix, we compile narrative evidence from early 2023 documenting how major U.S. companies began adjusting their hiring practices and internal workflows, explicitly attributing these changes to the adoption or anticipated impact of generative AI.

IBM: In May 2023, IBM’s CEO told *Bloomberg* that the company “*expects to pause hiring for roles as roughly 7,800 jobs could be replaced by Artificial Intelligence in the coming years*” (Reuters, 2023a).

Dropbox: In April 2023, Dropbox’s CEO announced that the company will reduce its workforce by 16 percent. According to his letter, one of the main explanations is that “*the AI era of computing has finally arrived. We’ve believed for many years that AI will give us new superpowers and completely transform knowledge work.*” (Houston, 2023).

Meta: In February 2023, Meta created a “*top-level product group*” to “*turbocharge*” generative-AI work and then announced 10,000 layoffs and cancellation of 5,000 open roles (March 2023) while emphasizing a pivot to AI (Reuters, 2023b,c).

CNET: In March 2023, shortly after revealing its use of AI-generated articles, CNET laid off about 10 percent of its staff. Parent company Red Ventures cited a refocus on search-optimized content and appointed the editor-in-chief to lead AI strategy, signaling ongoing integration of generative AI (Mia Sato, 2023).

Nuance and Microsoft: In March 2023, Nuance, a Microsoft subsidiary, launched *DAX Express*, a GPT-4–powered tool that automates clinical documentation from patient interactions, aiming to reduce physicians’ administrative workload and burnout (Medical Design & Development Staff, 2023).

Morgan Stanley: In March 2023, the Wealth Management division launched a new GPT-4–powered assistant. According to its co-president, the technology was meant to help “*freeing up valuable time for financial advisors to do what they do best—serve their clients*” (Morgan Stanley, 2023).

Wendy’s: In May 2023, Wendy’s partnered with Google Cloud to pilot *FreshAI*, a generative-AI system automating drive-thru orders using large language models to handle complex, customizable requests and improve accuracy and speed (The Wendy’s Company, 2023).

Presto and CKE Restaurants: Similarly, in May 2023, Presto Automation expanded its partnership with CKE Restaurants to deploy *Presto Voice*, an AI system automating drive-thru orders at Carl’s Jr. and Hardee’s. After pilots showed higher sales and efficiency, CKE began offering the technology to franchisees nationwide (TMCnet, 2023).

A.22 Most Common High- and Low-Exposed Occupations by Industry

Table A.6: Most Common Low/High-Exposure Occupations by Industry (Share of All ONET Roles in Industry)

NAICS	Low Exposure (Top 5)	High Exposure (Top 5)
3 <i>Manufacturing</i>	– Door-to-Door Sales Workers, News and Street Vendors, and Related Workers (5.4%) – Retail Salespersons (2.2%) – Maintenance Workers, Machinery (1.8%) – Machinists (1.4%) – Biofuels Processing Technicians (1.4%)	– Software Developers (6.8%) – Computer User Support Specialists (2.7%) – Customer Service Representatives (2.5%) – Bioengineers and Biomedical Engineers (2.0%) – Validation Engineers (1.9%)
4 <i>Trade / Retail</i>	– Retail Salespersons (9.9%) – Gambling Change Persons and Booth Cashiers (5.8%) – Cashiers (4.6%) – Stockers and Order Fillers (4.0%) – Merchandise Displayers and Window Trimmers (3.5%)	– Customer Service Representatives (5.8%) – Computer User Support Specialists (1.7%) – Bookkeeping, Accounting, and Auditing Clerks (1.6%) – Software Developers (1.6%) – Sales Representatives, Wholesale and Manufacturing, Except Technical and Scientific Products (1.6%)
51 <i>Information</i>	– Actors (2.3%) – Retail Salespersons (1.7%) – Door-to-Door Sales Workers, News and Street Vendors, and Related Workers (1.2%) – Nannies (1.1%) – Career/Technical Education Teachers, Secondary School (1.0%)	– Software Developers (13.1%) – Writers and Authors (6.0%) – News Analysts, Reporters, and Journalists (5.6%) – Editors (4.2%) – Customer Service Representatives (4.0%)
52 <i>Finance & Insurance</i>	– Gambling Change Persons and Booth Cashiers (0.9%) – Door-to-Door Sales Workers, News and Street Vendors, and Related Workers (0.8%) – Phlebotomists (0.6%) – Retail Salespersons (0.6%) – Nannies (0.5%)	– Loan Officers (7.9%) – Customer Service Representatives (7.2%) – Securities, Commodities, and Financial Services Sales Agents (6.1%) – Loan Interviewers and Clerks (5.5%) – Software Developers (4.4%)
54 <i>Professional Services</i>	– Demonstrators and Product Promoters (1.0%) – Medical and Clinical Laboratory Technicians (1.0%) – Door-to-Door Sales Workers, News and Street Vendors, and Related Workers (0.9%) – Retail Salespersons (0.9%) – Career/Technical Education Teachers, Secondary School (0.7%)	– Software Developers (7.4%) – Writers and Authors (5.7%) – Accountants and Auditors (4.1%) – Computer User Support Specialists (4.1%) – Public Relations Specialists (3.2%)
61 <i>Educational Services</i>	– Substitute Teachers, Short-Term (20.0%) – Career/Technical Education Teachers, Secondary School (19.1%) – Coaches and Scouts (3.8%) – Lifeguards, Ski Patrol, and Other Recreational Protective Service Workers (3.5%) – Nannies (3.4%)	– Public Relations Specialists (3.1%) – Computer User Support Specialists (2.6%) – Executive Secretaries and Executive Administrative Assistants (1.8%) – Software Developers (1.8%) – Writers and Authors (1.7%)
62 <i>Health Care & Social Assistance</i>	– Acute Care Nurses (14.3%) – Licensed Practical and Licensed Vocational Nurses (11.5%) – Phlebotomists (7.9%) – Home Health Aides (6.2%) – Nannies (5.4%)	– Eligibility Interviewers, Government Programs (2.2%) – Executive Secretaries and Executive Administrative Assistants (1.9%) – Public Relations Specialists (1.8%) – Computer User Support Specialists (1.7%) – Customer Service Representatives (1.5%)